Heterodox Finance: A First Introduction

Aggregation, uncertainty, endogenous risk, leverage, money and power

FED Course — Heterodox Finance, Lecture 0

What You Should Be Able to Do After This Lecture

1. Say what the mainstream (CAPM, EMH) and the heterodox views take as *given*, and what each tries to explain.
2. Explain the fallacy of composition and why the representative-agent shortcut hides it.
3. Distinguish risk from fundamental uncertainty, and say why this matters for μ, Σ and for market efficiency.
4. Describe endogenous risk and the feedback loop between measured volatility, leverage and prices.
5. Work through the leverage arithmetic, and describe Kalecki's principle of increasing risk and Minsky's hedge, speculative and Ponzi finance.
6. Explain why money is not a veil: endogenous money, liquidity preference, the hierarchy of money and the hierarchy of currencies.
7. Say where power enters finance, and in which ways finance and the real economy affect each other.
8. Use the 2007–08 crisis, the euro-area crisis and March 2020 to say what the CAPM and EMH can and cannot explain, and what the heterodox lens adds.

Where This Lecture Sits

In words first

So far the course has built the *mainstream* account of finance: investors choose portfolios (Markowitz, Tobin), prices adjust until markets clear (CAPM), and prices reflect available information (EMH). This lecture introduces a different way of thinking, for students who have not met it before. It starts with *methodological questions*, not equations.

	Mainstream block	**This lecture**
Static	Fisher, Markowitz, Tobin, CAPM, EMH	Which assumptions carry the weight?
Dynamic	Samuelson, Merton (intuition)	What if risk and money are not given?
Heterodox	**Lecture 0 (today)**	Keynes, Kalecki, Minsky and others

Outline

1. Two Ways of Looking at Finance
2. Seven Methodological Issues
3. Who's Who: The Main Contributors
4. Three Episodes as Tests
5. A Fair Assessment
6. Roadmap and Discussion Questions

Two Traditions, One Disagreement

In words first

Both traditions agree that finance matters and that prices, debts and returns are linked. They disagree about what to treat as *given*.

Mainstream

Beliefs, risks and the money system are *given*. The question is how prices adjust.

Heterodox

Where do beliefs, risks and money *come from*? They are part of what is to be explained.

Not rational versus irrational investors, but **what the model leaves out**.

Two Pictures: Agents, the Future, Risk, Debt

	Mainstream (CAPM, EMH)	**Heterodox (Keynes, Kalecki, Minsky)**
Agents	One representative investor, or many with the *same* beliefs	Many agents with different balance sheets, beliefs and power, who interact
The future	*Risk*: a known probability distribution that can be estimated	*Uncertainty*: no reliable probabilities for key events; decisions rest on convention and confidence
Risk	A given property of an asset (its σ and covariances)	Partly *created* by what investors collectively do
Debt and leverage	Financing is a veil (Modigliani–Miller)	Debt commitments shape investment and fragility; leverage moves with the cycle

Two Pictures: Money, Power, Dynamics

	Mainstream (CAPM, EMH)	**Heterodox (Keynes, Kalecki, Minsky)**
Money	A numeraire, plus a risk-free asset available to all at one rate	A hierarchy of promises: liquidity is scarce, and banks create money when they lend
Power	Everyone is a price-taker	Creditors, dealers, central banks and states set terms and decide who bears losses
Dynamics	Equilibrium; shocks come from outside	Cycles generated from within: stability breeds instability

Common misreading to avoid

Heterodox does *not* mean "investors are irrational". Investors can be perfectly sensible given what they know. The claim is that **sensible individual behaviour can have destabilizing collective consequences**, and that the world they face is not one in which the odds are known.

Key Terms in Plain Words

Fallacy of composition What is true for one agent is not true for all agents acting together.

Fundamental uncertainty Situations in which no reliable probabilities exist, because the future is not a repeat of the past.

Endogenous risk Risk generated by the behaviour of market participants themselves.

Leverage The ratio of assets to own capital: the more borrowed, the larger the effect of a given price change on own capital.

Liquidity The ease of turning an asset into money at a known price without moving it.

Minsky moment The point at which a long period of rising debt and rising prices turns into a collapse.

The Seven Issues at a Glance

1. **Aggregation**: from the one to the many
2. **Uncertainty**: risk is not the same as "not knowing"
3. **Endogenous risk**: risk is partly made by us
4. **Leverage**: how debt amplifies
5. **Money**: not a veil
6. **Power**: who sets the terms
7. **Finance and the real economy**: causation runs both ways

How to use them

Each issue names an assumption of the CAPM or EMH that the heterodox view questions. Keep a running list: by the end you should be able to match each CAPM assumption to a heterodox concept (Question 1).

1. Aggregation: from the One to the Many

In words first

One bank that sells assets to cut its risk is acting prudently. If *every* bank does it at once, prices fall, every bank's risk rises, and the prudence has made things worse. Adding up individuals is not the same as scaling up one individual.

The CAPM adds up by assumption

Everyone sees the same μ and Σ, so everyone holds the same risky portfolio T, and the market portfolio M is just T again. **The economy behaves like one investor.**

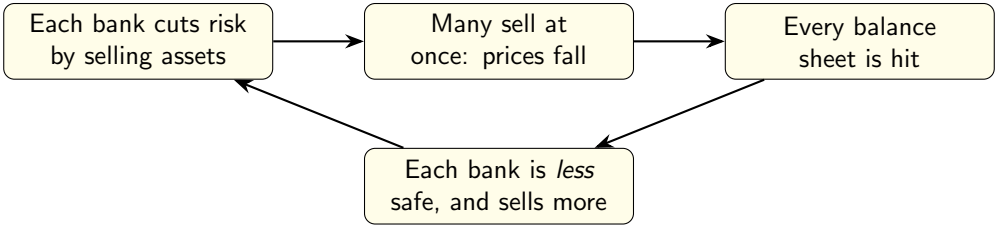
Three Heterodox Objections to Aggregation

1. **The representative agent is a modelling convenience, not a result.** Even if every individual is rational, there is no general reason for the aggregate to behave like one rational person. The Sonnenschein–Mantel–Debreu results show that aggregate demand can have almost any shape; Kirman (1992) argues that "the representative individual" represents no one.

2. **Fallacy of composition.** Keynes's paradox of thrift: if each household saves more, total demand falls, incomes fall, and saving may fall. The same logic applies to selling assets, cutting leverage or "de-risking".

3. **Aggregation hides debt.** In the CAPM the Bus is in zero net supply: every loan is someone else's debt, so in total they cancel. But *who* owes *whom* matters: a fall in asset prices ruins the borrowers while the lenders are paid in full, and the borrowers' distress spreads. Fisher (1933) called this *debt-deflation*.

Aggregation: Individual Prudence, Collective Fragility



A representative-agent model contains one bank, so this loop cannot even be written down. We return to it as the endogenous-risk loop (Issue 3) and the leverage spiral (Issue 4).

2. Uncertainty: Risk Is Not the Same as "Not Knowing"

In words first

A roulette wheel has *risk*: the odds are known. Whether a new technology will pay off in fifteen years, or whether a government will default, is different: no one can write down the odds. Keynes and Knight called the first *risk* and the second *uncertainty*.

Risk

Known distribution. Probabilities can be estimated from past data. Expected value is meaningful.

Fundamental uncertainty

No reliable probabilities. The future is not a repeat of the past. Decisions rest on convention and confidence.

Keynes (1937): "We Simply Do Not Know"

About the prospects of, for example, the price of copper and the rate of interest twenty years hence, "there is no scientific basis on which to form any calculable probability whatever. We simply do not know."

Investors still must act. Keynes argued that they do so by:

- **convention**: assume the present state of affairs will continue;
- **the opinion of others**: if everyone else is doing it, it is probably right;
- **animal spirits**: "spontaneous urge to action rather than inaction" (Keynes, 1936, ch. 12).

The beauty contest

Professional investors do not pick the asset they think is best, but the asset they think *others* will think best.

What Uncertainty Means for the Models We Have Studied

1. The inputs μ and Σ are not objective facts waiting to be estimated. "Homogeneous expectations" then has nothing to be homogeneous about.
2. The EMH says price equals the discounted expected payoff. With uncertainty there is no unique expectation, and a price can be driven by shifts in confidence with no change in "the fundamentals".
3. Davidson (1991) puts the same point in statistical terms: economic processes are *non-ergodic*, so averages over past data need not describe the future.

Link to the 25-sigma remark (Issue 3 and Section 4)

A model calibrated on a calm past can call a crisis a "25-sigma event". The heterodox reading: the model, not the world, was wrong.

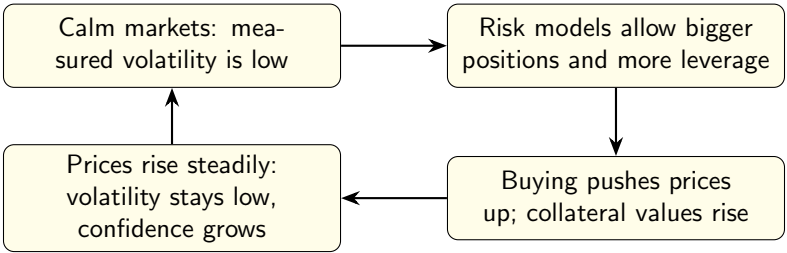
3. Endogenous Risk: Risk Is Partly Made by Us

In words first

The CAPM treats volatility and correlations as properties of assets, like their weight. In practice they depend on what investors are doing. When many hold the same positions and must sell together, volatility rises *because* they are selling.

- Danielsson and Shin (2003) call this *endogenous risk*.
- Minsky's slogan for the boom half of the cycle: "**stability is destabilizing**".
- A calm period convinces everyone that risk is low, and this leads them to take more.

The Endogenous-Risk Loop (Boom)



The loop strengthens itself.

The Same Loop in Reverse (Bust), and Why Risk Systems Feed It

Bust

A shock raises measured risk $\rightarrow$ models and lenders cut leverage $\rightarrow$ forced sales push prices down $\rightarrow$ volatility rises further.

Two features of real risk management feed the loop:

- **Value-at-risk limits** are computed from recent volatility, so low measured risk permits large positions (Adrian and Shin, 2014).
- **Market liquidity and funding liquidity reinforce each other**: when funding dries up, assets can only be sold at a discount (Brunnermeier and Pedersen, 2009).

Contrast with the CAPM

In the CAPM σ and ρ are fixed inputs. Here they are *outputs* of the behaviour of the people who use the model.

4. Leverage: How Debt Amplifies

In words first

With leverage, a small fall in the price of what you own can wipe out your own capital. If you then try to repair your balance sheet by selling, you push the price down again.

Example. A bank has assets of 100, financed by 95 of debt and 5 of equity:

$$\text{leverage} = \frac{\text{assets}}{\text{equity}} = \frac{100}{5} = 20.$$

Suppose asset prices fall by 2%.

The Arithmetic of a 2% Fall

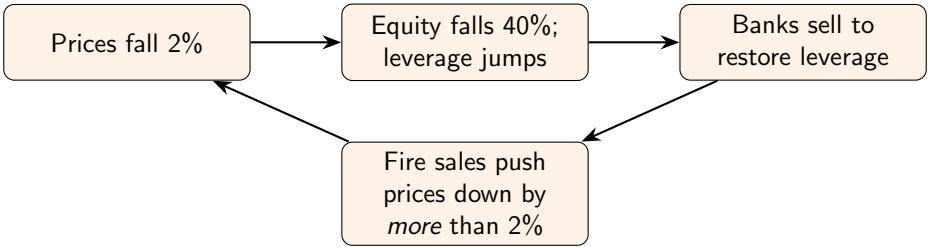
	Assets	Debt	Equity	Leverage
Before	100	95	5	20
After a 2% price fall	98	95	3	$98/3 = 32.7$
Back to leverage 20 (sell 38, repay debt)	60	57	3	20

- A 2% fall in prices cuts equity by 40%.
- To return to leverage 20 the bank must shrink assets from 98 to 60: it sells 38, almost 39% of its assets.

Procyclical Leverage and Fire Sales

Banks that *target* their leverage

They must sell when prices fall, and can buy when prices rise (Adrian and Shin, 2010). Leverage is therefore **procyclical**.



If many banks do this together, the fire sales push prices down by more than the original 2%, and the loop of Issue 3 runs.

Try It: Lower Leverage

Exercise (Question 3)

Redo the calculation for a bank with leverage 10 and a 2% price fall.

1. What is the new leverage?
2. How much must it sell to restore leverage 10?
3. Why does lower leverage make the system less fragile?

Set-up: assets 100, equity 10, debt 90. Fall of 2% takes assets to 98. Work out equity, new leverage, target assets, and the sale.

Answer: Leverage 10 versus Leverage 20

	Assets	Debt	Equity	Leverage
Before	100	90	10	10
After a 2% price fall	98	90	8	$98/8 = 12.25$
Back to leverage 10 (sell 18, repay debt)	80	72	8	10

	Leverage 20	**Leverage 10**
Fall in equity from a 2% price fall	40%	20%
Forced sale to restore leverage (out of 98)	38 ($\approx 39\%$)	18 ($\approx 18\%$)

Lower leverage means a smaller loss of equity and a smaller forced sale, so less downward pressure on prices if many banks are in the same position.

Kalecki's Principle of Increasing Risk (1937)

In words first

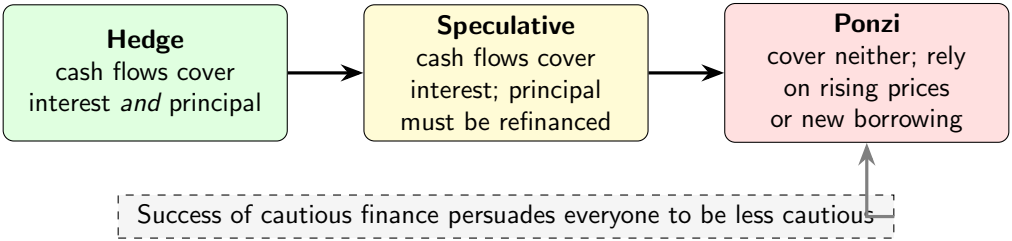
The more you borrow relative to your own capital, the more you and your lender stand to lose if things go wrong.

- The greater an investor's borrowing relative to her own capital, the greater the risk that the *lender* bears and the risk that she *herself* bears.
- Lenders therefore charge more, or ration, as leverage rises.
- A rising borrowing rate breaks the CAPM's single risk-free rate, and makes the choice of leverage *itself* part of the risk.

Contrast with Tobin

In Tobin's world, anyone can borrow at r_f without limit. Here the borrowing rate rises with the amount borrowed.

Minsky's Three Financing Postures (1986)



In a long boom the economy drifts from hedge towards speculative and Ponzi finance. **Fragility is built up in good times.**

Leverage: What to Take Away

- Leverage turns a small price change into a large change in own capital.
- Banks that target leverage sell into falling markets: **leverage is procyclical**, and collective selling is a fallacy of composition.
- Kalecki: the cost of borrowing rises with leverage, so there is no single risk-free rate.
- Minsky: the *composition* of finance (hedge, speculative, Ponzi) shifts towards fragility during booms.

Mainstream benchmark

In Modigliani–Miller's world none of this matters for real investment: financing is a veil. We return to this under Money.

5. Money: Not a Veil

In words first

In the CAPM there is a bank that lends and borrows at one rate, without limit, to everyone. Real money is a ladder of promises, and in a crisis everyone tries to climb to the top rung at once.

Six ideas on money:

1. Banks create money
2. Liquidity preference
3. A hierarchy of money
4. A hierarchy of currencies
5. The state and the value of money
6. Modigliani–Miller: finance as a veil

Banks Create Money

- When a bank makes a loan it *creates a deposit*. It does not first collect deposits and then lend them out (McLeay, Radia and Thomas, 2014).
- The supply of credit therefore depends on banks' willingness to lend and on borrowers' expectations, *not* on a fixed pool of saving.

Why this matters for the CAPM

In the CAPM, saving is the fixed resource that gets allocated across assets. If credit is created by banks, then the amount of leverage in the system moves with confidence: **finance is a driver of the cycle, not a mirror of it**.

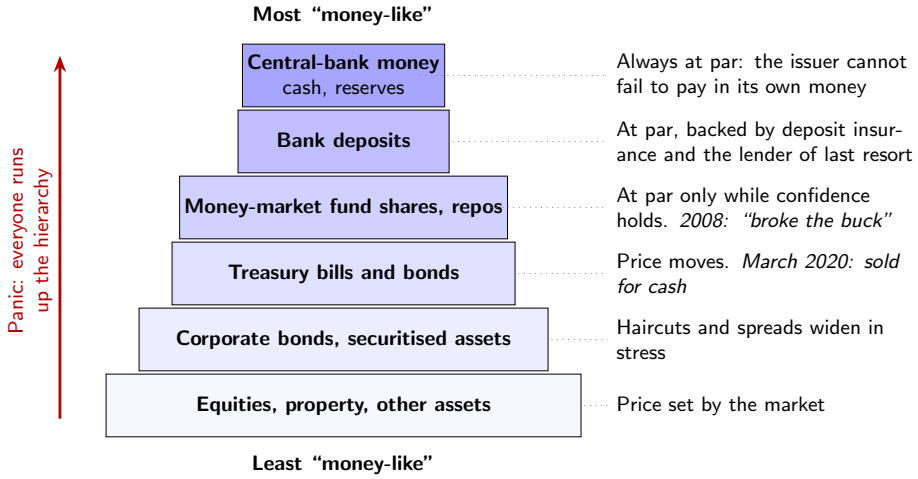
Liquidity Preference

- Keynes asked why anyone holds money that earns no interest.
- The answer is **uncertainty**: money gives the freedom to wait.
- The more uncertain the future, the stronger the preference for money, and the lower the demand for risky assets.

Link to Issue 2

Under risk, holding a zero-return asset is dominated. Under uncertainty it is a sensible response. This is why the demand for safe assets can jump in a crisis with no change in "fundamentals".

A Hierarchy of Money (after Mehrling, 2011)



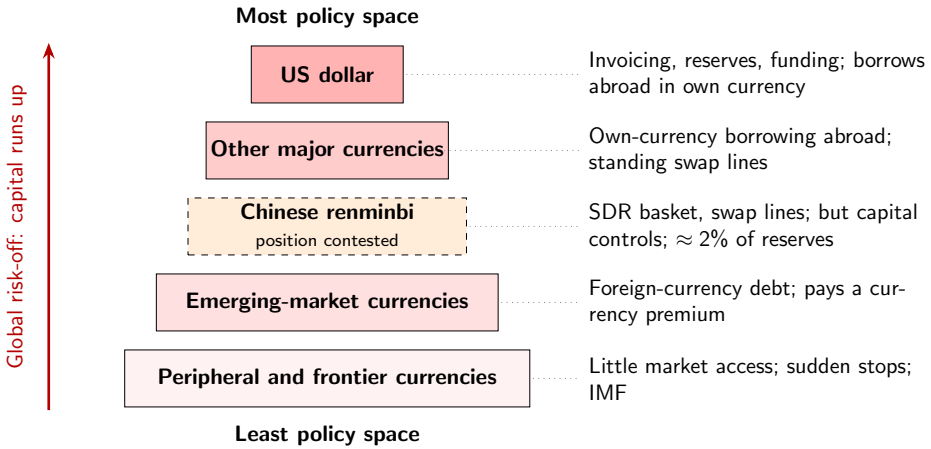
Reading the Hierarchy

- Each layer *promises* convertibility into the layer above at par.
- In calm markets the promise is credible, and the rungs look interchangeable.
- In a panic the promise weakens as one moves down, and holders run *up* the hierarchy.
- The lender of last resort extends the top layer's guarantee downward, and **decides how far**.

Implication for the CAPM

The CAPM's "risk-free asset" is treated as a property of an asset. In the money view, an asset's safety depends on *where it sits* in the hierarchy and on who backs it.

A Hierarchy of Currencies



Currency Hierarchy: Features and Global Risk-off

Tier	In normal times	In a global risk-off
US dollar	Used worldwide for invoicing, reserves and funding. Borrows abroad in its own currency.	Safe-haven demand rises. Fed swap lines to 14 central banks, March 2020.
Other majors (euro, yen, sterling, Swiss franc)	Deep markets. Borrows abroad in own currency. Standing Fed swap lines (ECB, BoJ, BoE, SNB, BoC).	Backstopped. Yen and franc often act as safe havens.
Chinese renminbi	Second-largest economy, SDR basket, 32 bilateral swap lines (2025). Capital controls limit use: about 2% of reserves (2025Q4).	Swap lines are bilateral, not a global backstop; not yet a safe haven.
Emerging-market (e.g. real, peso, rupee)	Shallower markets. Foreign debt often in foreign currency. Pays a currency premium.	Outflows, depreciation, reserves sold. Only a few got temporary swap lines.
Peripheral and frontier	Little market access. Foreign-currency borrowing dominates.	Sudden stops and balance-of-payments crises; reliance on the IMF.

Examples are illustrative; placement of a single currency is a matter of degree and debate.

Why the Hierarchy Matters

- Currencies are not equal (Cohen, 1998; Kaltenbrunner, 2015; Fritz, de Paula and Prates, 2018).
- A country whose currency is low in the hierarchy:
  - borrows abroad largely in foreign currency;
  - pays a premium to hold its own currency;
  - sees capital flow out when global risk appetite falls (Eichengreen and Hausmann, 1999; Rey, 2015).
- Its policy space is narrower, and the **balance-of-payments constraint binds earlier**.
- The hierarchy is *self-reinforcing*: low-ranked currencies are held less, and so remain low-ranked.

The renminbi: the contested case

Second-largest economy, SDR basket, a network of bilateral swap lines; but capital controls and limited convertibility keep its share of official reserves at about 2% (IMF, 2026; People's Bank of China, 2025).

The State and the Value of Money

- **State theory of money** (Knapp, 1924; Lerner, 1943): money has value, in part, because the state demands taxes in it and stands behind its banks.
- **Hahn's problem** (previous note): with a known last date, backward induction implies that intrinsically worthless money has no value in any period.

The link

A finite-horizon argument cannot unravel money whose value rests on *institutions*: the state and its tax claims. This is one answer to Hahn.

Modigliani–Miller: Finance as a Veil

Mainstream benchmark

Modigliani and Miller (1958): the way a firm is financed does not change its value. A sound benchmark, but it implies that credit and money *cannot* affect real investment.

Heterodox view

The opposite: credit conditions are among the main drivers of investment and asset prices. Debt commitments, liquidity and the hierarchy of money all matter.

Modigliani–Miller is a useful *benchmark*: it tells you what has to be added (frictions, uncertainty, power) for finance to matter.

6. Power: Who Sets the Terms

In words first

The CAPM assumes that every investor is a price-taker and that every borrower can borrow at the same rate. In reality some actors set the terms on which others can borrow, trade, and survive.

- **Creditors and debtors.** Lenders decide on rates, collateral and whether to roll over a loan. In a crisis, "who takes the loss" is settled by bargaining, law and politics, not by the pricing model.
- **Business confidence as a veto.** Kalecki (1943): when governments depend on private investment, the confidence of business and the financial markets gives them a powerful say over policy.

Power: Central Banks, States and Financialisation

- **Central banks and states.** A lender of last resort decides which institutions and which markets are rescued, and on what terms. Such decisions determine who bears the costs of financial excess.
- **Financialisation.** Since the 1980s the weight of financial markets and the interests of shareholders in corporate decisions has grown. Heterodox economists argue that this has shifted income towards finance and away from investment and wages (Stockhammer, 2004; Epstein, 2005).

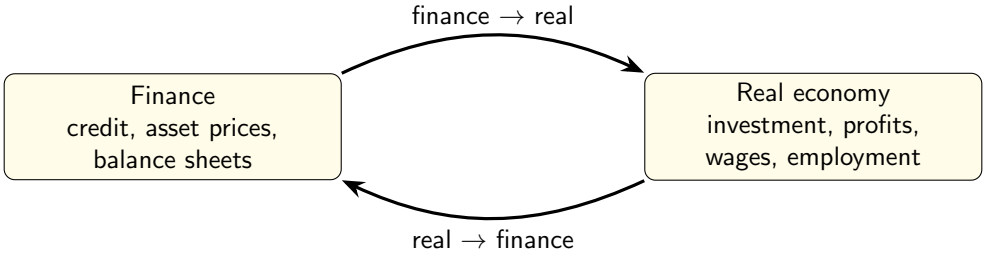
Where we will see power again

The Irish bank guarantee, the ECB's 2012 promise, and the Fed's swap lines in March 2020 are all decisions about who is protected, and on what terms.

7. Finance and the Real Economy: Causation Runs Both Ways

In words first

In the mainstream picture, real cash flows come first, and finance only prices them. In the heterodox picture, finance also shapes the real economy, and the real economy feeds back on finance.



Two-Way Causation: the Mechanisms

	Finance → real variables	**Real variables → finance**
Mechanism	Credit supply and asset prices determine investment, demand and employment; the terms of finance depend on the balance sheets of lenders and borrowers	Profits and wages determine whether debts can be paid; weak demand leads to defaults and losses
Kalecki	Investment, financed by credit, creates the profits that validate it: "workers spend what they earn, capitalists earn what they spend"	Profits are the source of debt service; falling profits make past debts unsustainable
Minsky	The financing posture of firms determines how fragile investment is	A downturn turns hedge into speculative and Ponzi units
Mainstream benchmark	Modigliani–Miller: financing does not matter	Cash flows are given; prices follow

Inequality, Household Debt, and Debt as a Social Relation

- A related heterodox thread links rising inequality to household borrowing: families whose incomes stagnate borrow to sustain consumption (Barba and Pivetti, 2009).
- The point is the same as for firms and banks: **debt is a social relation between those who pay and those who receive**.
- Who owes whom, and who has the power to demand repayment, determines how a downturn is distributed.

Seven issues, one theme

Aggregation, uncertainty, endogenous risk, leverage, money, power, and two-way causation each remove a "given" of the mainstream model: finance is part of the system it is supposed to price.

Who's Who (1): Keynes, Kalecki, Minsky, Fisher

Thinker	Core idea	Main source
J.M. Keynes	Fundamental uncertainty; convention and animal spirits; liquidity preference; the casino warning ("When the capital development of a country becomes a by-product of the activities of a casino, the job is likely to be ill-done")	*Treatise on Probability* (1921); *General Theory* (1936), ch. 12; QJE (1937)
M. Kalecki	Principle of increasing risk; profits come from capitalists' own spending; power of business confidence over policy	Economica (1937); Political Quarterly (1943)
H. Minsky	Financial instability hypothesis; hedge, speculative and Ponzi finance; "stability is destabilizing"	*Stabilizing an Unstable Economy* (1986)
I. Fisher	Debt-deflation: falling prices raise the real burden of debt, leading to more distress selling	Econometrica (1933)

Who's Who (2): Kindleberger, Godley, Davidson, Mehrling

Thinker	Core idea	Main source
C. Kindleberger	Manias, panics and crashes follow a common pattern; the need for a lender of last resort	*Manias, Panics and Crashes* (1978)
W. Godley	Stock-flow consistent modelling: every flow is someone's income and someone's expenditure, every asset is someone's liability; warning on the euro design	"Maastricht and all that" (1992)
P. Davidson	Non-ergodicity: the future is not a statistical average of the past	JEP (1991)
P. Mehrling	The money view: a hierarchy of money; dealers and central banks as "market makers of last resort"	*The New Lombard Street* (2011)

This list is a starting point. Brock and Hommes (heterogeneous traders), Soros (reflexivity) and others are taken up in the technical notes.

Which Thinker for Which Issue?

Issue	**Main voices**
Aggregation	Keynes (paradox of thrift), Fisher (debt-deflation), Kirman
Uncertainty	Keynes, Davidson
Endogenous risk	Minsky, Danielsson and Shin, Brunnermeier and Pedersen
Leverage	Kalecki, Minsky, Adrian and Shin
Money	Keynes, Knapp, Lerner, Mehrling, Cohen, Kaltenbrunner
Power	Kalecki, Kindleberger, Epstein, Stockhammer
Finance and the real economy	Kalecki, Minsky, Godley, Barba and Pivetti

Using Episodes as Tests

In words first

A theory is tested by whether it helps us understand what happened. We take three episodes and ask the same two questions of each: *what do the CAPM and EMH tell us, and what is missing?*

We keep to the facts that are well documented, and say where interpretations differ.

1. The 2007–08 crisis
2. The euro-area crisis
3. March 2020: the dash for cash

The Three Episodes Side by Side

	2007–08 crisis	**Euro-area crisis**	**March 2020**
Mainstream reading	New information about defaults repriced assets; risk models were mis-specified	Markets repriced sovereign risk as fundamentals became known	Pandemic news repriced all risky assets
What it cannot explain	Models of calm risk; runs on "safe" assets; why prices fell far below fundamentals	Near-identical yields before 2008; yields moving by two points on a speech	The safe asset being sold; correlations rising towards one
Heterodox concepts	Leverage, Minsky postures, endogenous risk, run on privately created money	Money-issuer power, convention, bank–sovereign loop, who bears losses	Liquidity preference, hierarchy of money, dollar swap lines

2007–08: Facts (1) The Build-up

- A long housing boom, financed by mortgage lending that moved towards ever riskier borrowers.
- Loans were packaged into securities and funded, to a large extent, by *short-term wholesale borrowing*: repurchase agreements and asset-backed commercial paper.
- Large US investment banks held assets of the order of **thirty times their equity**.

Link to the leverage arithmetic

At leverage 30, a fall of about 3.3% in asset values wipes out equity: $1/30 \approx 3.3\%$.

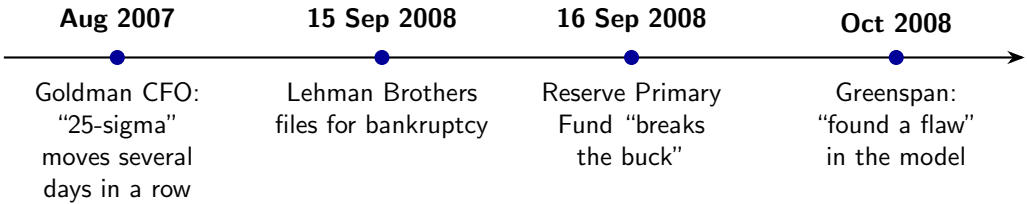
2007–08: Facts (2) The "25-Sigma" Moves

- In August 2007 the chief financial officer of Goldman Sachs said that the firm had seen "25-standard-deviation" moves several days in a row.
- Under a normal distribution a 25σ event has a probability of about 3×10^{-138}.

What the number tells us

The explanation is not bad luck. It is a model that does not describe the world: a calm sample period, fixed σ and ρ, no feedback from selling to volatility. This is **endogenous risk** and **uncertainty**, not a rare draw from a known distribution.

2007–08: Timeline of the Run



2007–08: Facts (3) The Run on Money-Market Funds

- The Reserve Primary Fund, a money-market fund of about $62 billion, held $785 million of Lehman debt, and announced that it would "break the buck".
- Shareholders asked for about $40 billion within two days.
- In the week of 15 September roughly $300 billion, or 14%, left prime money-market funds.
- The US Treasury temporarily guaranteed more than $3 trillion of money-market fund shares, and the Federal Reserve lent to banks so that they could buy assets from the funds (Schapiro, 2012).

In October 2008 Alan Greenspan told Congress that he had "found a flaw" in the model that he perceived to be "the critical functioning structure that defines how the world works".

2007–08: What the CAPM and EMH Struggle With

1. **Known, stable risk.** Value-at-risk systems extrapolated a calm past. The diversification across regional housing markets that the models relied on disappeared because house prices fell together: *a systematic risk had been priced as idiosyncratic*.
2. **A safe asset.** The CAPM needs a safe asset available to all. Money-market funds were treated as cash-like, and the run on them was a run on a privately created form of money.
3. **Exogenous shocks.** Volatility did not rise in proportion to news about fundamentals. It rose as leveraged institutions sold into falling markets: an endogenous-risk spiral.

2007–08: The Heterodox Lens

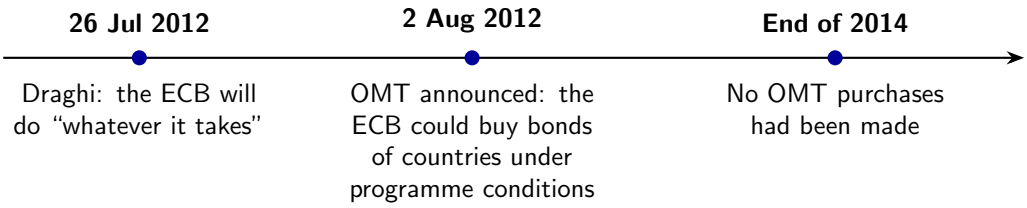
- **Minsky:** borrowers whose payments depended on refinancing against rising house prices were, in effect, in speculative or Ponzi positions; the stability of the preceding years raised the willingness to lend.
- **Leverage and shadow banking** supplied the transmission (procyclical leverage, run on repo).
- **Hierarchy of money:** the lender of last resort, at the top, stopped the run.
- **Power:** who was rescued, on what terms, and who bore the losses.

See Brunnermeier (2009), Gorton and Metrick (2012), Wray (2008).

Euro Area: Facts (1) Convergence and Divergence

- In the years before 2008, interest rates on the government bonds of Greece, Ireland, Portugal, Spain and Italy converged to those of Germany, *as if default risk had vanished*.
- After 2009 spreads widened sharply, and by 2011–12 the Italian and Spanish ten-year yields exceeded 7%.
- In September 2008 the Irish government guaranteed the liabilities of six banks: €440 billion, more than twice the country's annual GDP. Bank liabilities became a potential sovereign burden.

Euro Area: Facts (2) "Whatever It Takes"



An ECB working paper found that the announcement lowered two-year Italian and Spanish yields by about **two percentage points**, while German and French yields were largely unchanged.

Euro Area: What the CAPM and EMH Struggle With

1. If prices reflect fundamentals, why were countries with very different public finances priced almost alike before 2008?
2. Why did yields move by two points on a speech, with *no purchases and no change in the debt*?
3. The "risk-free" status of a government bond is not a property of the bond. It depends on whether a central bank stands behind it.

Compare with the CAPM

The CAPM's risk-free rate r_f is a given. Here it is created by a policy commitment.

Euro Area: The Heterodox Lens

- **Currency issuer or user.** Euro-area members use a currency they do not issue. Unlike countries with their own central bank, their governments can face a liquidity run that becomes a solvency problem (Godley, 1992; De Grauwe and Ji, 2013).
- **Money and power:** the credible promise of a lender of last resort changed the price *without being exercised*.
- **Convention:** pre-crisis convergence is consistent with Keynes's account of convention.
- **Aggregation:** the bank–sovereign loop in Ireland shows how an aggregation of private debts onto the state can make sovereign risk endogenous.
- **Power:** who was guaranteed, and who paid for it.

March 2020: Facts (1) From Flight to Quality to Dash for Cash

- Late February 2020: flight to safe government bonds; yields fell. Between the start of 2020 and 9 March the US ten-year yield fell by about 137 basis points.
- From about 10 March it reversed: ten-year yields in the US, UK and Germany jumped by around 60 basis points.
- The flight to quality had become "a global dash for cash": investors sold whatever could be sold to meet margin calls, redemptions and the need for cash.
- Trading conditions deteriorated most for US Treasuries.

March 2020: Facts (2) Sellers and Dealers

- Foreign official institutions reduced their Treasury holdings by about $150 billion in March.
- Mutual funds sold to meet redemptions; leveraged funds unwound basis trades, a sign of rapid deleveraging.
- Dealers' Treasury inventories had tripled between 2017 and the eve of the crisis, limiting their capacity to absorb sales (Barone et al., 2023).

The policy response

The Federal Reserve responded with large-scale purchases and, on 19 March, opened temporary dollar swap lines with nine more central banks, joining the five with standing lines: **fourteen in all**. Drawings on the lines peaked at $449 billion in late May.

March 2020: What the CAPM and EMH Struggle With

- The risk-free asset is supposed to be the *safe haven*. In March 2020 it was **sold**.
- The CAPM's diversification argument relies on bonds and stocks *not* falling together. In the dash for cash, they both fell.
- Pandemic news explains a fall in equity prices, but *not* a fall in the price of the safest bond.

Correlation is not a constant

The CAPM treats ρ as a parameter. In a dash for cash, correlations rise towards one because everyone sells whatever can be sold.

March 2020: The Heterodox Lens

- **Liquidity preference in its pure form:** with uncertainty at its peak, everyone wanted the top of the hierarchy of money (Mehrling, 2011).
- **Dealers' balance sheets are limited:** they could not be the buyers of last resort, so the central bank had to become one.
- **Hierarchy of currencies:** the issuer of the dollar decides which central banks, and therefore which banks abroad, get access. Only fourteen did. *This is power exercised through the money system.*
- **Two-way causation:** a real shock (a pandemic) was transmitted and magnified through leverage and liquidity, and in turn required fiscal and monetary support for the real economy.

The EMH Is Not Refuted by a Fall in Prices on Bad News

What the mainstream can say

If prices fell because the news was bad, an efficient market could do exactly this.

What the episodes challenge is:

- the *mechanism* (endogenous risk, forced selling);
- the stability of the assumed risk estimates;
- the idea of a unique "fundamental" value.

This is the joint-hypothesis problem again: a model of prices and a model of risk are tested together.

Mainstream Finance Has Responded

- **Limits to arbitrage** (Shleifer and Vishny, 1997).
- **Funding-liquidity spirals** (Brunnermeier and Pedersen, 2009).
- **Intermediary asset pricing** (He and Krishnamurthy, 2013).

These put leverage and balance sheets *inside* the model.

The heterodox response

Heterodox economists welcome this, and say it treats as a "friction" what Minsky and Kalecki treated as *central*.

The Heterodox Approach Has Costs

- It is less often expressed in a single estimable model.
- It is more difficult to forecast with.

Where its strengths lie

Explaining *why* crises happen, *how* they spread, and *who* pays.

	Mainstream	**Heterodox**
Strength	Tractable, testable, forecasts	Explains instability, distribution, power
Weakness	Treats risk, money and balance sheets as given	Less often one estimable model

Where to Go Next

- *Finance Is Not Outside the Model* (Kalecki–Minsky companion): leverage with a rising borrowing rate, endogenous risk, Minsky's fallacy of composition.
- *Heterodox Finance (1)*: the portfolio problem restated by Kalecki, Minsky and Brock–Hommes.
- *Heterodox Finance (2)*: CAPM, efficient markets and the endogenous return.
- The CAPM and EMH heterodox audits: a step-by-step audit of each assumption.

Discussion Questions

1. List the five CAPM assumptions. For each, name the heterodox concept of this lecture that challenges it.

2. A risk manager says the 2007 losses were "a 25-sigma event". What does this say about the world, and what does it say about the model?

3. In the leverage example, redo the calculation for a bank with leverage 10 and a 2% price fall. How much must it sell to restore leverage 10? Why does lower leverage make the system less fragile? *(Answer: slide "Leverage 10 versus Leverage 20".)*

4. In July–August 2012 yields fell sharply on a speech and a programme that was never used. What does this tell us about the "fundamental" price of sovereign debt?

5. In March 2020, which asset was the true "risk-free" asset, and who decided? Explain using the hierarchy of money.

Summary: Seven Issues, Three Tests

Issue	**Mainstream assumption questioned**	**Where it showed up**
Aggregation	Representative agent; all hold T	Fire sales; Irish bank–sovereign loop
Uncertainty	Known μ, Σ; same beliefs	"25-sigma" moves; yields on a speech
Endogenous risk	σ and ρ are given	Volatility spirals in 2008 and 2020
Leverage	Financing is a veil; one borrowing rate	Leverage near thirty; basis-trade unwinding
Money	A risk-free asset for all	Run on money-market funds; Treasuries sold
Power	Price-taking	Guarantees, OMT, swap lines
Real–finance link	Cash flows first, prices follow	Housing boom and bust; pandemic support

Take-Home Messages

1. The two traditions differ in what they treat as *given*: beliefs, risks and money.
2. Sensible individual behaviour can have destabilizing collective consequences (**fallacy of composition**).
3. **Uncertainty** is not risk: μ and Σ are not objective facts.
4. Risk is partly **endogenous**, and **leverage** is procyclical.
5. **Money** is a hierarchy of promises, and so are currencies.
6. **Power** decides who sets terms and who bears losses.
7. Finance and the real economy affect each other.
8. The EMH is not refuted by a fall in prices on bad news, but the *mechanisms* of the three episodes are hard to fit.

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